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ANALYSIS OF USA AIR TRAFFIC AND FLIGHT DATA

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ANALYSIS OF USA FLIGHT DATA

INTRODUCTION

The following report analyses data of flights within the USA. The analysis and visualizations included focus primarily on the data collected in the year 2006, whilst the 'airport' and 'plane-data' datasets were also used as supplementary datasets to aid in the analytical process. All datasets were obtained from the 2009 ASA Statistical Computing and Graphics Data Expo.

The year 2006 was chosen as it contains approximately 7.5 million data entries, thereby allowing for sufficient room to perform data-wrangling and preprocessing, whilst maintaining the quality of the data.

WHAT ARE THE BEST TIMES, AND DAYS OF THE WEEK TO MINIMIZE DELAYS

Data Pre-processing

The columns of most value, when analyzing the occurrence of delays, are the arrival delay and departure delay columns. Hence all the null and missing values within these columns were imputed with their respective mean values in-order to avoid data-loss. Furthermore, the correlation coefficient between arrival delays and departure delays was discovered to be approximately 0.9 indicating a high correlation. Hence it was appropriate to introduce a 'Total Delay' column containing the summation of the Arrival and Departure delays per flight.

All entries where a delay does not occur were then filtered out, thus allowing for the analysis of only flights that were delayed. The 'CRSDepTime' column was used to calculate the hour in which the flight departed, with a new column being added that indicated the hour of departure of the flight (each day was segmented into 24 hours).

Additionally, in-order to calculate the best days of the week where delays are a minimum, the data was segmented into days of the week and a box plot was plotted using the appropriate data in order to get a proper understanding of the distribution of delay times.

Analysis

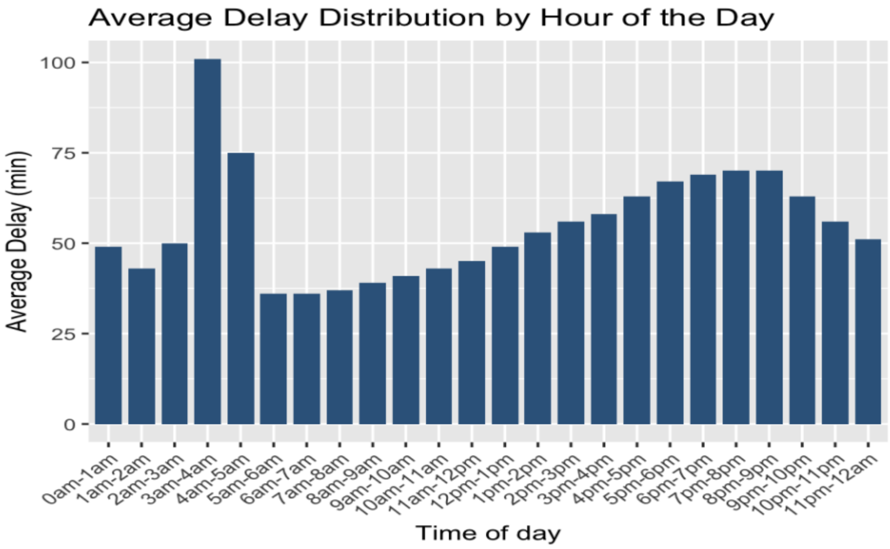


Fig 3. In R

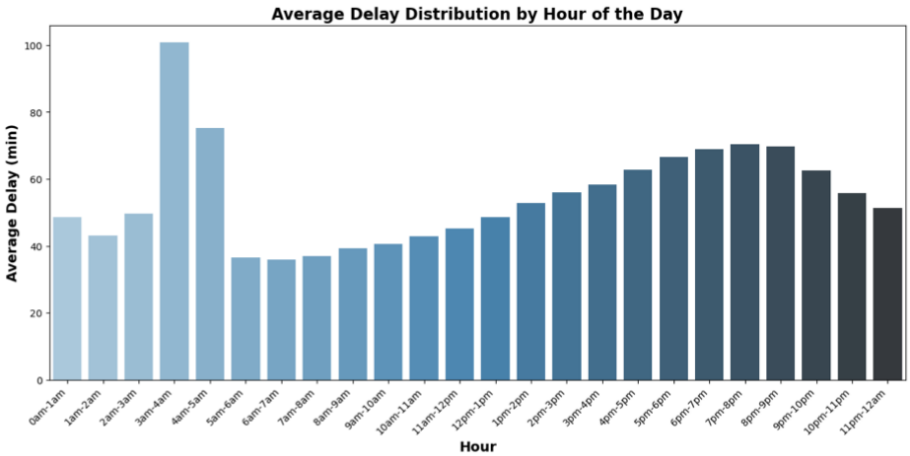


Fig 3. In Python

The average delay of a flight departed during each hour of the day was calculated.

When plotted as a bar-chart (Fig 3) it is clear that the highest average delay occurs for flights departed between 3 a.m. – 4 a.m. (An average of approximately 100 minutes – i.e. one hour and 40 minutes). Furthermore, delays are minimized in the morning hours (5 a.m. – 11 a.m.). There is once again a sharp increase in the average delay as the day progresses with delay time being maximum during 7 p.m. to 8 p.m. for the evening time period.

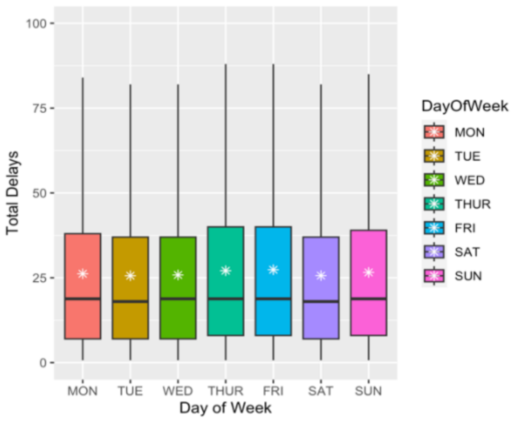


*Fig 4. In R (mean is indicated by *)*

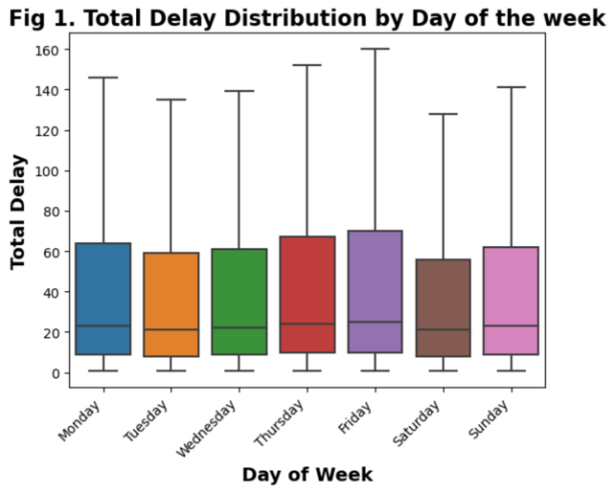


Fig 4. In Python

As displayed by the box-plots given above, there does not appear to be a large deviation between the average delays pertaining to each day of the week. Tuesday and Saturday do appear to have a lower spread of delay times, while also having relatively lower mean delays, and are thereby more favorable choices, if wanting to minimize delays.

Monday, Friday and Sunday however, appear to have a higher mean when it comes to delays. This is to be expected as most travelling is done by passengers at the start and end of the week, thus leaving room for larger delay times for flights.

DO OLDER PLANES SUFFER MORE DELAYS

Data Pre-processing

The use of the 'plane-data' dataset was vital in answering the above question as it provided information pertaining to the year in which a particular plane started its service. This information is valuable in determining the age of the plane as per the day the data is being recorded. It is worth keeping in mind that the '2006' dataset is still being used. Once again, the means of the 'Arrival Delay' and 'Departure Delay' columns were imputed in place of the null and missing values in the respective columns so as to avoid data loss. A 'Total Delays' Variable was also calculated as per the previous part. Furthermore, as it would not be appropriate to impute the start of service year of a plane, planes having no start of service year were dropped from the dataset. Upon merging of the main dataset with the 'airplane' dataset, the age of the plane, per record was calculated and assigned as a new variable/

column. Records where age is less than ‘0’ were dropped as this would indicate that the plane has started service in the years following 2006, which is obviously an error in the data entry process. Also, data-points where a delay does not occur were filtered out from the final dataset. The average ‘Total delay’ per plane age was then calculated in-order to determine if older planes tend to suffer more delays than relatively newer ones.

Analysis

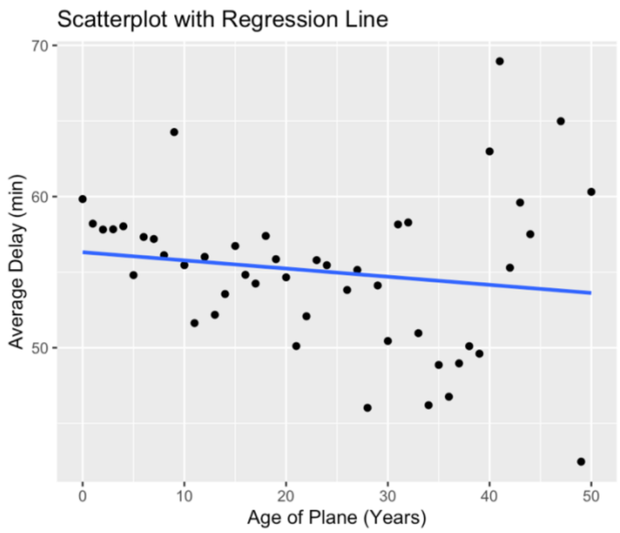


Fig 5. In R

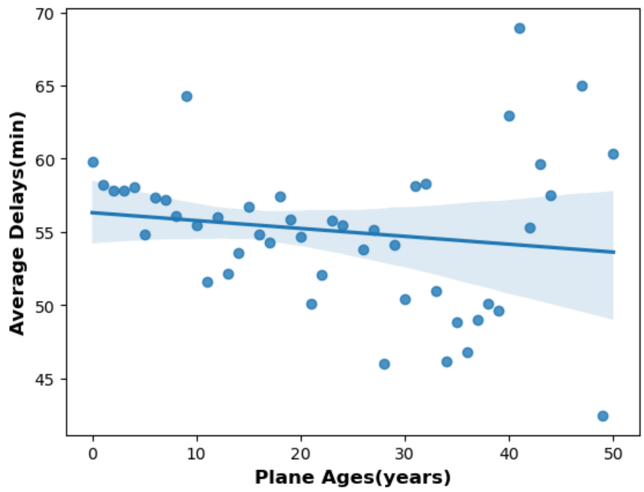


Fig 5. In Python

When a scatter-plot relating plane-age to their respective average ‘Total Delay’ times is plotted, it somewhat surprisingly shows that there is a slight weak negative correlation (coefficient = -0.152) between a planes age and its average delay time. This implies that older planes tend to suffer from lesser delay times than newer planes. The outcome may well be somewhat skewed given the fact that older planes do not fly as often as newer planes, do not fly commercially and are most likely to carry very few passengers over relatively short distances.

In order to further explore this, plane ages were grouped into classes with 10-year intervals, and boxplots were drawn for each class.

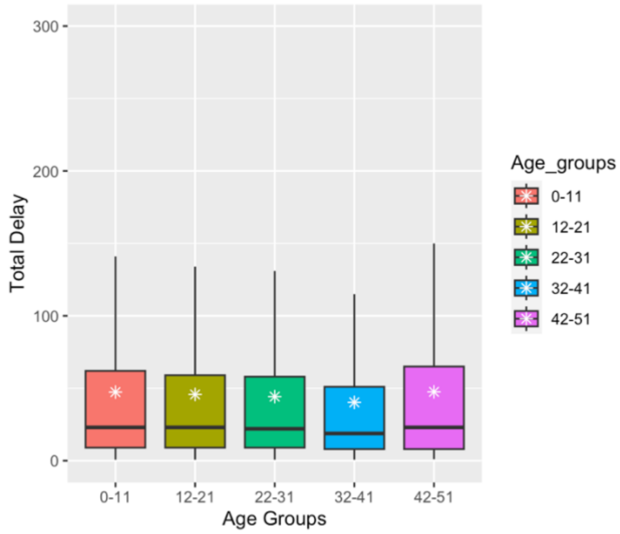


*Fig 6. In R (mean is indicated by *)*

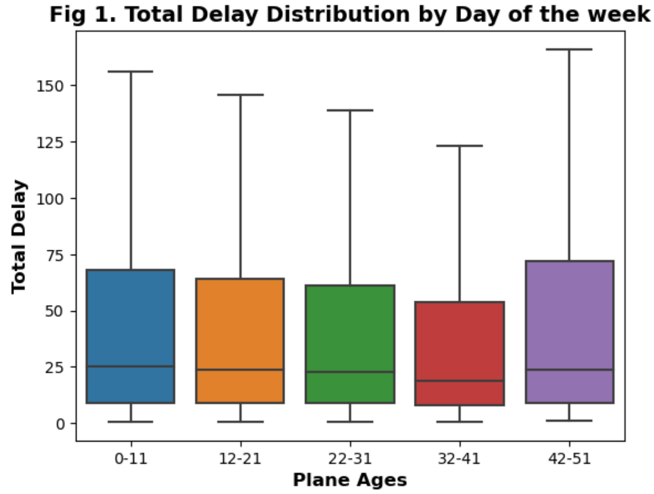


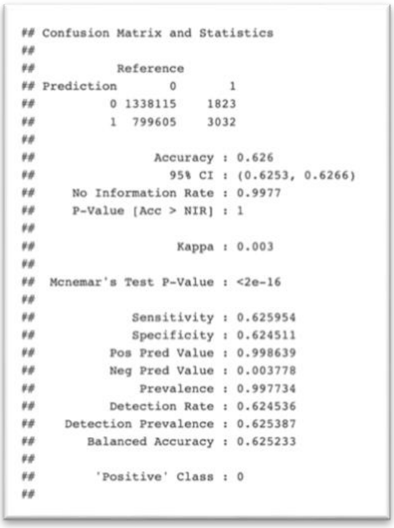
Fig 6. In Python

It is evident from the plot given above that when grouped by ages, on average, planes belonging to the (42–51) year-old group tend to have a higher average delay. Yet the (32-41) year-old age group has less of a spread of Total delay times and a relatively lower mean. Therefore, while keeping in mind the fact that the Planes' Age and Average Total delay have a weak correlation, we can reach a conclusion that older planes do not tend to suffer more delays than newer planes.

FOR EACH YEAR FIT A LOGISTIC REGRESSION MODEL FOR THE PROBABILITY OF DIVERTED US FLIGHTS AND VISUALIZE THE COEFFITIENTS (ONLY THE YEAR 2006 WAS USED)

Data Pre-processing

The 'airports' dataset was used in-order to get information about the longitude and latitude of arrival and departure airports. In order to fit a logistic regression model to predict the probability of diverted US flights, a total of 10 features were used. Due to the presence of a large class imbalance in the response variable, it was necessary to resample (random under-sampling was used) the train data. All null and missing values in numeric features were imputed with either their mean or median, and categorical features were subject to encoding. Numeric features were also scaled using standard scaling. In python, a pipeline was implemented to streamline the above process. The following confusion matrix and report were obtained.



```
## Confusion Matrix and Statistics
##
##           Reference
## Prediction       0       1
##          0 1338115    1823
##          1  799605    3032
##
##                Accuracy : 0.626
##                  95% CI : (0.6253, 0.6266)
##     No Information Rate : 0.9977
##     P-Value [Acc > NIR] : 1
##
##                   Kappa : 0.003
##
##  Mcnemar's Test P-Value : <2e-16
##
##             Sensitivity : 0.625954
##             Specificity : 0.624511
##          Pos Pred Value : 0.998639
##          Neg Pred Value : 0.003778
##              Prevalence : 0.997734
##          Detection Rate : 0.624536
##    Detection Prevalence : 0.625387
##       Balanced Accuracy : 0.625233
##
##        'Positive' Class : 0
##
```

Fig 7. In R

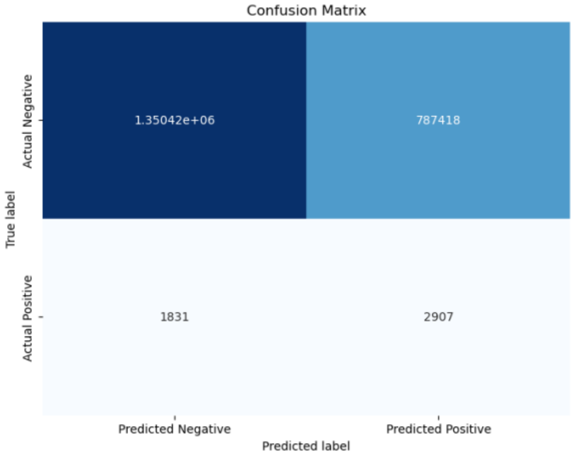


Fig 7. In Python

VISUALIZING THE COEFFITIENTS

The coefficients obtained by the model could be visualized as per fig 8.

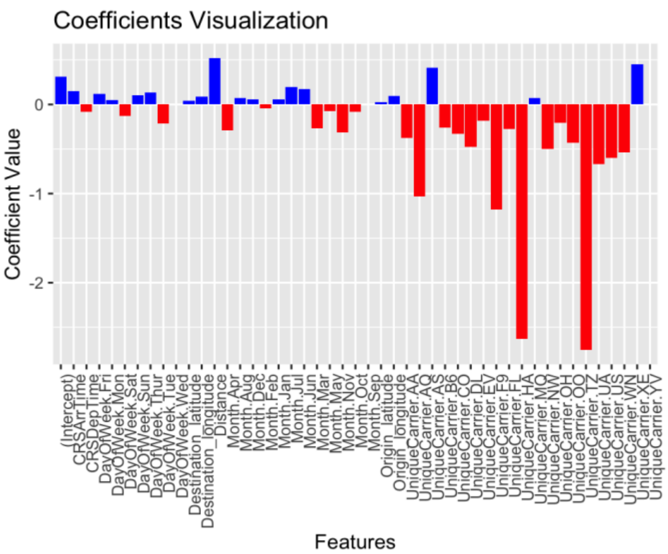


Fig 8. In R

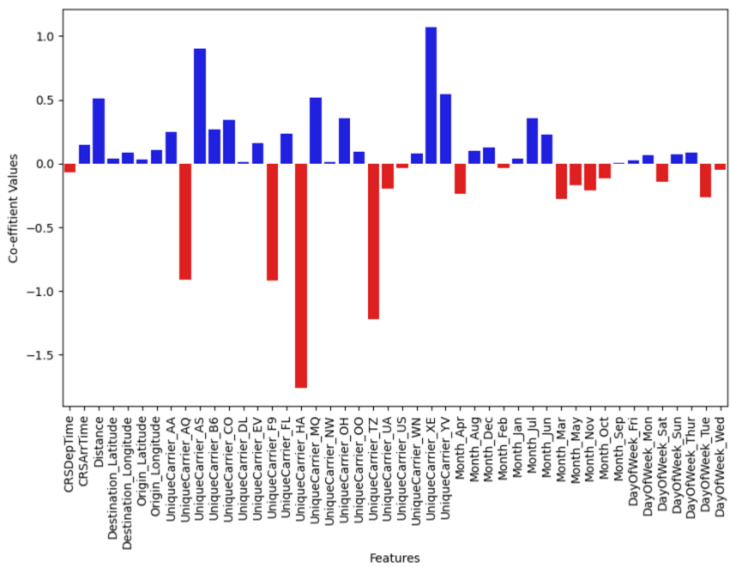


Fig 8. In Python